Pseudocode for algorithms

Algorithm 1: RelaxedDO algorithm

Input: Graph, update
Output: Incremental count, $tc + \frac{tc2}{2}$
$tc \leftarrow 0, tc2 \leftarrow 0$;
for $u \in B_{u,t}$ **do**
  $H \leftarrow N_{u,t}$;
  for $v \in N^+_{u,t}, w \in N_{v,t} \wedge (v < w) \wedge (w \in H)$ **do**
    tc2++;
  end
  for $v \in N_{u,t}, w \in N^+_{v,t} \wedge (v < w) \wedge (w \in H)$ **do**
    tc2++;
  end
  for $v \in N^+_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++
  end
  $H \leftarrow N^+_{u,t}$;
  for $v \in N_{u,t}, w \in N_{v,t} \wedge (v < w) \wedge (w \in H)$ **do**
    tc2++;
  end
  for $v \in N_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N^+_{u,t}, w \in N_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N^+_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++
  end
end

Algorithm 2: RepeatOld algorithm

Input: Graph, update
Output: Incremental count, tc
$tc \leftarrow 0$;
$B_{v,t} \leftarrow \{\forall u \in V \mid (u,v) \in V \times V \wedge (u,v) \in B\}$;
$V_{ad} \leftarrow \{\forall v \in V \mid (v,w) \in V \times V \wedge w \in B_{v,t} \wedge v \notin B_{v,t}\}$;
for $u \in B_{u,t}$ **do**
  $H \leftarrow N_{u,t}$;
  for $v \in N^+_{u,t}, w \in N_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N^+_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  $H \leftarrow N^+_{u,t}$;
  for $v \in N_{u,t}, w \in N_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N^+_{u,t}, w \in N_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
  for $v \in N^+_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
end
for $u \in V_{ad}$ **do**
  $H \leftarrow N_{u,t}$;
  for $v \in N_{u,t}, w \in N^+_{v,t} \wedge (u < v) \wedge (v < w) \wedge (w \in H)$ **do**
    tc++;
  end
end